ORD CLEARANCE FORM

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Product Title

Summertime methane and carbon dioxide emission rates and associated variables from a national-scale survey of 146 reservoirs in the United States

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Impact / Purpose Statement

Note: The Impact / Purpose Statement information for this work product will be displayed on the additional pages.

Product Description / Abstract

Note: All Product Description / Abstract information for this work product will be displayed on the additional pages.

Tracking and Planning

Note: All Tracking and Planning Field data for this work product will be displayed on the additional pages.

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Quality Assurance	Accepted Date	Available Online Date	Published Date
Is there an approved QAPP (or QAPPs) supporting this product? ☑ Yes ☐ No ☐ Not Applicable	11/10/2025	03/13/2026	03/13/2026

QAPP Reference
J-WECD-0032592-QP-1-12

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2. reservoirs	5.
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Additional Digital Signatures (As applicable) - Extra digital signatures may be displayed on the next page.	
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Sub-Product ID and Title
ACE.401.2.4.19: Summertime methane and carbon dioxide emission rates and associated variables from a national-scale survey of 146 reservoirs in the United States

Tracking and Planning 2019 Forward Field Set(s)
Research Area ID: ACE.401
Research Area: Sources and Sinks of Air Pollution and Climate Forcers

Product Title: Improved Estimates of GHG Emissions from Waterbodies for the US Inventory of GHG Emissions and Sinks.

Brief Description and Use: The Office of Air and Radiation (OAR) fulfills US Treaty obligations under the 1992 United Nations Framework Convention on Climate Change (UNFCCC) by preparing the annual Inventory of U.S. Greenhouse Gas (GHG) Emissions and Sinks. In 2019, the Intergovernmental Panel on Climate Change (IPCC) published guidance, coauthored by ORD, for including emissions from surface waters in the Inventory. Until the guidance was released, surface waters were the largest anthropogenic methane source not included in the Inventory. ORD used IPCC default emission factors (EFs) to estimate emissions from surface waters in the 1990 ¿ 2020 US Inventory, but it isn¿t clear how well the default EFs reflect conditions in the US. For example, the data used to derive the default EFs for ponds and canals include only three US waterbodies. Extrapolating from these sparse measurements to the conterminous US has resulted in a poorly constrained emission estimate for these systems. The accuracy of the default EFs for US reservoirs is also uncertain. An ORD study demonstrated that EFs can vary by four orders of magnitude across reservoirs within a single US state, highlighting the potential bias that could result from using a default EF. To improve the accuracy of EFs for US surface waters, ORD will execute measurements and conduct a literature review. New measurements include the SuRGE project (Survey of Reservoir Greenhouse gas Emissions), a national-scale survey of reservoir EFs. This work was planned as a three-year effort under StRAP3, but was delayed due to COVID complications. The final sampling season is proposed to occur during the first year of StRAP4. Data from SuRGE will be used to train a model for predicting EFs at unsampled locations, thereby generating site-specific EFs for all US reservoirs. During years 2-4 of StRAP4, ORD will lead a national-scale field-survey of EFs from US ponds. During StRAP4 ORD will also conduct a literature review to locate available data on EFs for US canals and ditches. Another source of uncertainty in the Inventory is the spatial extent of managed waterbodies in the US. Under StRAP4, ORD will utilize datasets containing information on the distribution, management, and size of US waterbodies to improve the geospatial representation of managed US waterbodies. Results from the efforts described above will be integrated into the 1990-2025 Inventory. This Product will use country specific EFs and a consistent geospatial representation of US managed waterbodies to generate an accurate and well constrained estimate of anthropogenic GHG emissions from US surface waters.

Topic(s):
Understanding Air Pollution and Climate Change and Their Impacts on Human Health and Ecosystems

Research Program Area: Air, Climate, and Energy

Impact / Purpose Statement
This article & dataset presents a rich collection of physicochemical parameters from 147 reservoirs distributed across the conterminous U.S. One hundred and eight of the reservoirs were selected using a statistical survey design and can provide unbiased inferences to the condition of all U.S. reservoirs. These data could be of interest to local water management specialists or those assessing the ecological condition of reservoirs at the national scale.

Product Description / Abstract
Reservoirs are globally important sources of greenhouse gases, but the magnitude of their emissions is highly uncertain. Here, we present data for 146 reservoirs from two surveys of reservoir methane and carbon dioxide emissions, one at the regional scale in the midwestern United States and one at the national scale in the United States, plus data from two hand-picked sites in Washington and Puerto Rico. At all reservoirs, ebullitive and diffusive emissions and basic physicochemistry were measured at 15–55 locations during one 22 to 64-h period during the summers of 2016–2023, with four reservoirs revisited a second time. Contemporaneous water chemistry measurements were made at one or two locations in each reservoir. The dataset consists of two geospatial files and seven CSV files containing greenhouse gas emissions, water chemistry, morphology, and other relevant data. To date, these data comprise the largest multi-reservoir emissions dataset assembled using consistent measurement methods.

Does this journal article have data associated with it?
Yes: EPA Data

Data Description:
Primary/secondary data 'owned' by EPA through in-house or EPA-funded efforts (ScienceHub full entry)

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Sent to PubMed Central?
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PubMed Central Embargo Period:
12 months

Comments

Author: Margie Vazquez   Date: 06/12/2025 2:22 PM   System Source: STICS
This manuscript and dataset are well covered by the cited QAPP. SuRGE project QAPPs were updated regularly, sometimes more than once per sampling season. QA was well covered, including an addendum examining any possible affects to the data from COVID-caused delays of analyses. The manuscript and supplemental information track very well with the plans in the QAPP - from sampling design to target analyses. This QA product review (filed under QA Track # J-WECD-0032592-JA-1-0)is limited to confirmation that research presented in a product being cleared is covered by an approved QAPP as per EPA requirements.

Author: Matthew Heberling   Date: 06/17/2025 7:36 AM   System Source: STICS
I have reviewed the paper and SI material as acting supervisor. The paper is well written and I found no major issues. I have some minor suggestions that I will provide Jake on hardcopy (i.e., typos, disclaimer twice?, acronyms not defined). Based on the minor revisions, I think this paper can move forward and I will approve.

Author: Susan Cormier   Date: 06/17/2025 10:52 AM   System Source: STICS
No technical or policy issues. The best documented data set that I have seen. Well done!

Author: Gayle Hagler   Date: 06/24/2025 10:11 PM   System Source: STICS
This is a very well done paper that I anticipate will be highly cited. There are a few comments for consideration - please see the version with "GH" at the end of the file name. If you have any questions, please reach out and happy to talk through these comments. Since this is a dataset release paper without policy connections drawn, I plan to share for general awareness with IOAA and recommend against a formal AN process.

Author: Gayle Hagler   Date: 06/26/2025 3:32 PM   System Source: STICS
Returning as we discussed in a call - please update with final version when you are ready and technical review form (or equivalent) documentation of the code review for the associated paper's code on GitHub.

Author: Leeann Young   Date: 07/16/2025 8:00 AM   System Source: STICS
Track changes (Beaulieu Paper LOL Reservoir 20250715-GH-JB.docx) and clean (Beaulieu Paper Clean LOL Reservoir 20250715-GH-JB.docx) versions of manuscript reflecting revisions following Gayle Hagler’s review. Also uploading technical review form for code review.

Author: Gayle Hagler   Date: 07/23/2025 7:29 AM   System Source: STICS
This is good to go forwards with journal submission, including release of the cleared code on GitHub via a public repository. IOAA AN process is not necessary given the scope of the paper. I may share the clean version for awareness purposes to IOAA.